

Homework Section 12.5

1. Find parametric equations of the line:
 - a) Through the point $(2, 0, -5)$ and parallel to $\mathbf{v} = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$.
 - b) Through the point $(-1, 3, 2)$ and parallel to the line $x = 1 + 2t$, $y = 3t$ and $z = 4 - 3t$.
2.
 - a) Find a vector equation for the line in 1(a).
 - b) Find the symmetric equations of the line from 1(b).
3. Find the points where the line $x = 1 + 2t$, $y = 3t$, and $z = 4 - 3t$ intersects the yz -plane.
4. Decide if each pair of lines are intersecting, skew, or parallel. If there is an intersection point, find it.
 - a) Line 1: $x = 1 + 2t$, $y = 4 + 3t$ and $z = 4 - 5t$
Line 2: $x = 2 - 8s$, $y = -12s$ and $z = -11 + 20s$
 - b) Line 1: $x = 2 + 2t$, $y = 4 + t$ and $z = 4 - 5t$
Line 2: $x = -8s$, $y = -1 - 3s$ and $z = 4 + s$
 - c) Line 1: $x = 2 + 2t$, $y = 4 + t$ and $z = 11 - t$
Line 2: $x = -8s$, $y = -1 - 3s$ and $z = 7s$
5. Find an equation of the plane:
 - a) passing through $(-4, 1, 3)$ and perpendicular to the vector $\mathbf{v} = \langle 2, -7, 3 \rangle$.
 - b) passing through $(-4, 1, 3)$ and perpendicular to the line $x = 1 + 2t$, $y = 3t$ and $z = 4 - 3t$.
 - c) passing through the points $(-4, 1, 3)$, $(5, -1, -3)$, and $(2, 2, -6)$.
 - d) passing through $(-4, 1, 3)$ and containing the line $x = 1 + 2t$, $y = 3t$ and $z = 4 - 3t$.
6. Determine the point of intersection of the line $x = 2 + 2t$, $y = 4 + t$ and $z = 4 - 5t$ and the plane $x + 2y - 3z = 36$.
7. Find a set of parametric equations of the line of intersection of the planes $x - y + z = 1$ and $2x + y + z = 0$.
8. Decide if the planes below are perpendicular or parallel to the plane $2x - 9y + z = 4$.
 - a) $-6x + 27y - 3z = 9$
 - b) $-3x - y - 3z = -12$

9. Find the angle between the planes $x - y + z = 1$ and $2x + y + z = 0$. Round your answer to the nearest hundredth of a degree.
10. Find the distance between the parallel planes $2x - 9y + z = 4$ and $2x - 9y + z = 10$
11. Let the numerical entries in the below table represent a 5×3 matrix A . The table gives the points scored, number of rebounds, and number of minutes played for the starting five players of a basketball team in a single game. Find product $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \end{bmatrix} A$ and discuss what it represents.

Pts	Reb	Min	Player
21	5	35	Hurd
18	9	32	Spencer
10	7	28	Smith
8	3	30	Gomez
15	12	32	Johnson

Note: As this problem illustrates, matrices can be used to keep track of and manipulate large amounts of data.

12. A vector can be written as a column matrix or row matrix. In this class, when multiplying a matrix times a vector, the vector is usually written as either a 2×1 or 3×1 column matrix (often called a **column vector**). Perform the following matrix multiplications by hand:

a) $\begin{bmatrix} -1 & 3 \\ 2 & -5 \end{bmatrix} \begin{bmatrix} -4 \\ 10 \end{bmatrix}$

b) $\begin{bmatrix} 5 & 0 & -9 \\ 1 & -1 & 2 \\ 3 & 4 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$

Note: The product in part (a) is a 2×1 column vector, and the product in part (b) is 3×1 column vector.

Big Idea: Multiplying an $n \times n$ matrix times an $n \times 1$ column vector produces an $n \times 1$ column vector (for our purposes, n is either equal to 2 or 3). For example the multiplication in part (b) above *transforms* the vector $\langle 2, -1, 3 \rangle$ into the vector $\langle -17, 9, 2 \rangle$